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Methods for forming structures with desired crystallinity for MRAM applications

Abstract

Embodiments of the disclosure provide methods and apparatus for fabricating magnetic tunnel junction (MTJ) structures on a substrate in for spin-transfer-torque magnetoresistive random access memory (STT-MRAM) applications. In one example, a film stack utilized to form a magnetic tunnel junction structure on a substrate includes a pinned layer disposed on a substrate, wherein the pinned layer comprises multiple layers including at least one or more of a Co containing layer, Pt containing layer, Ta containing layer, an Ru containing layer, an optional structure decoupling layer disposed on the pinned magnetic layer, a magnetic reference layer disposed on the optional structure decoupling layer, a tunneling barrier layer disposed on the magnetic reference layer, a magnetic storage layer disposed on the tunneling barrier layer, and a capping layer disposed on the magnetic storage layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/438,420, filed Feb. 21, 2017, which is a continuation of U.S. patent application Ser. No. 15/199,006, filed Jun. 30, 2016, which claims benefit of U.S. Provisional Application Ser. No. 62/192,565 filed Jul. 14, 2015, which is incorporated by reference in its entirety.

FIELD

(1) Embodiments of the disclosure relate to methods for fabricating structures used in magnetoresistive random access memory (MRAM) applications. More specifically, embodiments of the disclosure relate to methods for fabricating magnetic tunnel junction structures for MRAM applications.

BACKGROUND

(2) Magnetoresistive random access memory (MRAM) is a type of memory device containing an array of MRAM cells that store data using their resistance values instead of electronic charges. Generally, each MRAM cell includes a magnetic tunnel junction (MTJ) structure. The MTJ structure may have adjustable resistance to represent a logic state “0” or “1.” The MTJ structure typically includes a stack of magnetic layers having a configuration in which two ferromagnetic layers are separated by a thin non-magnetic dielectric, e.g., an insulating tunneling layer. A top electrode and a bottom electrode are utilized to sandwich the MTJ structure so electric current may flow between the top and the bottom electrode.

(3) One ferromagnetic layer, e.g., a reference layer, is characterized by a magnetization with a fixed direction. The other ferromagnetic layer, e.g., a storage layer, is characterized by a magnetization with a direction that is varied upon writing of the device, such as by applying a magnetic field. In some devices, an insulator material, such as a dielectric oxide layer, may be formed as a thin tunneling barrier layer sandwiched between the ferromagnetic layers. The layers are typically deposited sequentially as overlying blanketed films. The ferromagnetic layers and the insulator material are subsequently patterned by various etching processes in which one or more layers are removed, either partially or totally, in order to form a device feature.

(4) When the respective magnetizations of the reference layer and the storage layer are antiparallel, a resistance of the magnetic tunnel junction is high having a resistance value $R_{sub,max}$ corresponding to a high logic state “1”. On the other hand, when the respective magnetizations are parallel, the resistance of the magnetic tunnel junction is low, namely having a resistance value $R_{sub,min}$ corresponding to a low logic state “0”. A logic state of a MRAM cell is read by comparing its resistance value to a reference resistance value $R_{sub,ref}$, which is derived from a reference cell or a group of reference cells and represents an in-between resistance value between that of the high logic state “1” and the low logic state “0”.

(5) One type of MRAM cell is spin-transfer-torque magnetic random access memory (STT-MRAM). While manufacturing conventional STT-MRAM devices, a thermal annealing process is often performed right after the film layer deposition process to assist crystallization of the ferromagnetic layers as well as the insulator material sandwiched in the device structure. Insufficient thermal energy or inaccurate temperature control during the annealing process may cause the film bonding structures or properties formed in an undesirable manner. For example, inaccurate temperature control or undesired drift of the thermal diffusion during the annealing process may result in insufficient crystallization of the film layer, leading to failure of the device to meet its intended performance.

(6) Therefore, there is a need in the art for improved methods and apparatus for fabricating MTJ structures for MRAM applications.

SUMMARY

(7) Embodiments of the disclosure provide methods and apparatus for fabricating magnetic tunnel junction (MTJ) structures on a substrate in MRAM applications, particularly for spin-transfer-torque magnetic random access memory (STT-MRAM) applications. In one embodiment, a film stack utilized to form a magnetic tunnel junction structure on a substrate includes a pinned layer disposed on a substrate, wherein the pinned layer comprises multiple layers including at least one or more of a Co containing layer, a Pt containing layer, a Ta containing layer, an Ru containing layer and combinations thereof, an optional structure decoupling layer disposed on the pinned magnetic layer, a magnetic reference layer disposed on the optional structure decoupling layer, a tunneling barrier layer disposed on the magnetic reference layer, a magnetic storage layer disposed on the tunneling barrier layer, and a capping layer disposed on the magnetic storage layer.

(8) In another embodiment, the method includes patterning a film stack having a tunneling barrier layer disposed between a magnetic reference layer and a magnetic storage layer disposed on a substrate to remove a portion of the film stack from the substrate until an upper surface of the

substrate is exposed, forming a sidewall passivation layer on sidewalls of the patterned film stack and subsequently performing a thermal annealing process to the film stack.

(9) In yet another embodiment, a method of forming a magnetic tunnel junction structure on a substrate includes patterning a film stack having a tunneling barrier layer disposed between a magnetic reference layer and a magnetic storage layer disposed on a substrate to remove a portion of the film stack from the substrate until an upper surface of the substrate is exposed, forming a sidewall passivation layer on sidewalls of the patterned film stack, forming an insulation layer in contact with the sidewall passivation layer on the substrate, and driving dopants from the magnetic reference layer and the magnetic storage layer laterally outward into the sidewall passivation layer by a thermal treatment process.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings.

(2) FIG. 1 depicts one embodiment of a plasma processing chamber for practice one embodiment of the present disclosure;

(3) FIG. 2 depicts one embodiment of an annealing chamber for practice one embodiment of the present disclosure;

(4) FIG. 3 depicts a flow diagram illustrating a method for fabricating magnetic tunnel junction (MTJ) structures according to one embodiment of the present disclosure;

(5) FIGS. 4A-4E are schematic side views of a substrate at various stages of the method of FIG. 3; and

(6) FIG. 5 is a schematic diagram of another embodiment of a portion of the film stack illustrated in FIG. 4A.

(7) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

(8) It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

DETAILED DESCRIPTION

(9) Embodiments of the disclosure generally provide apparatus and methods for forming MTJ structures from a film stack disposed on a substrate for MRAM applications. The apparatus and methods include forming the film properties of material layers from the film stack with desired crystallinity by a thermal process performed after the patterning of the material layers in the film stack. While patterning, a sidewall passivation layer may be formed along sidewalls of the film stack. Thus, by performing the thermal process after the patterning process and the sidewall passivation deposition process, the thermal energy provided from the thermal process may efficiently diffuse dopants into the material layers laterally outward toward the sidewall passivation layer, rather than vertically to adjacent material layers formed in the film stack. By doing so, a controllable degree of crystallinity of the material layers in the film stack may be obtained without damage or interference from the dopants that might potentially be cross-diffused from the adjacent materials. The film stack as patterned and annealed with desired degree of crystallinity may be used to form MTJ structures with desired dimension and features for MRAM applications.

(10) FIG. 1 is a sectional view of one example of a processing chamber **100** suitable for performing

a patterning process to etch a film stack disposed on a substrate. Suitable processing chambers that may be adapted for use with the teachings disclosed herein include, for example, an ENABLER®, C3® or AdvantEdge Mesa® processing chamber available from Applied Materials, Inc. of Santa Clara, California. Although the processing chamber **100** is shown including a plurality of features that enable superior etching performance, it is contemplated that other processing chambers may be adapted to benefit from one or more of the inventive features disclosed herein.

(11) The processing chamber **100** includes a chamber body **102** and a lid **104** which enclose an interior volume **106**. The chamber body **102** is typically fabricated from aluminum, stainless steel or other suitable material. The chamber body **102** generally includes sidewalls **108** and a bottom **110**. A substrate support pedestal access port (not shown) is generally defined in a sidewall **108** and is selectively sealed by a slit valve to facilitate entry and egress of a substrate **103** from the processing chamber **100**. An exhaust port **126** is defined in the chamber body **102** and couples the interior volume **106** to a pump system **128**. The pump system **128** generally includes one or more pumps and throttle valves utilized to evacuate and regulate the pressure of the interior volume **106** of the processing chamber **100**. In one implementation, the pump system **128** maintains the pressure inside the interior volume **106** at operating pressures typically between about 10 mTorr to about 500 Torr.

(12) The lid **104** is sealingly supported on the sidewall **108** of the chamber body **102**. The lid **104** may be opened to allow access to the interior volume **106** of the processing chamber **100**. The lid **104** includes a window **142** that facilitates optical process monitoring. In one implementation, the window **142** is comprised of quartz or other suitable material that is transmissive to a signal utilized by an optical monitoring system **140** mounted outside the processing chamber **100**.

(13) The optical monitoring system **140** is positioned to view at least one of the interior volume **106** of the chamber body **102** and/or the substrate **103** positioned on a substrate support pedestal assembly **148** through the window **142**. In one embodiment, the optical monitoring system **140** is coupled to the lid **104** and facilitates an integrated deposition process that uses optical metrology to provide information that enables process adjustment to compensate for incoming substrate pattern feature inconsistencies (such as thickness, and the like), provide process state monitoring (such as plasma monitoring, temperature monitoring, and the like) as needed. One optical monitoring system that may be adapted to benefit from the disclosure is the EyeD® full-spectrum, interferometric metrology module, available from Applied Materials, Inc., of Santa Clara, California.

(14) A gas panel **158** is coupled to the processing chamber **100** to provide process and/or cleaning gases to the interior volume **106**. In the example depicted in FIG. 1, inlet ports **132'**, **132''** are provided in the lid **104** to allow gases to be delivered from the gas panel **158** to the interior volume **106** of the processing chamber **100**. In one implementation, the gas panel **158** is adapted to provide fluorinated process gas through the inlet ports **132'**, **132''** and into the interior volume **106** of the processing chamber **100**. In one implementation, the process gas provided from the gas panel **158** includes at least a fluorinated gas, chlorine, and a carbon containing gas, an oxygen gas, a nitrogen containing gas and a chlorine containing gas. Examples of fluorinated and carbon containing gases include CHF₃, CH₂F₂ and CF₄. Other fluorinated gases may include one or more of C₂F₆, C₄F₆, C₃F₈ and C₅F₈. Examples of the oxygen containing gas include O₂, CO₂, CO, N₂O, NO₂, O₃, H₂O, and the like. Examples of the nitrogen containing gas include N₂, NH₃, N₂O, NO₂ and the like. Examples of the chlorine containing gas include HCl, Cl₂, CCl₄, CHCl₃, CH₂Cl₂, CH₃Cl, and the like. Suitable examples of the carbon containing gas include methane (CH₄), ethane (C₂H₆), ethylene (C₂H₄), and the like.

(15) A showerhead assembly **130** is coupled to an interior surface **114** of the lid **104**. The showerhead assembly **130** includes a plurality of apertures that allow the gases flowing through the showerhead assembly **130** from the inlet ports **132'**, **132''** into the interior volume **106** of the

processing chamber **100** in a predefined distribution across the surface of the substrate **103** being processed in the processing chamber **100**.

(16) A remote plasma source **177** may be optionally coupled to the gas panel **158** to facilitate dissociating gas mixture from a remote plasma prior to entering into the interior volume **106** for processing. A RF source power **143** is coupled through a matching network **141** to the showerhead assembly **130**. The RF source power **143** typically is capable of producing up to about 3000 W at a tunable frequency in a range from about 50 kHz to about 200 MHz.

(17) The showerhead assembly **130** additionally includes a region transmissive to an optical metrology signal. The optically transmissive region or passage **138** is suitable for allowing the optical monitoring system **140** to view the interior volume **106** and/or the substrate **103** positioned on the substrate support pedestal assembly **148**. The passage **138** may be a material, an aperture or plurality of apertures formed or disposed in the showerhead assembly **130** that is substantially transmissive to the wavelengths of energy generated by, and reflected back to, the optical monitoring system **140**. In one embodiment, the passage **138** includes a window **142** to prevent gas leakage through the passage **138**. The window **142** may be a sapphire plate, quartz plate or other suitable material. The window **142** may alternatively be disposed in the lid **104**.

(18) In one implementation, the showerhead assembly **130** is configured with a plurality of zones that allow for separate control of gas flowing into the interior volume **106** of the processing chamber **100**. In the example illustrated in FIG. 1, the showerhead assembly **130** as an inner zone **134** and an outer zone **136** that are separately coupled to the gas panel **158** through separate inlet ports **132'**, **132''**.

(19) The substrate support pedestal assembly **148** is disposed in the interior volume **106** of the processing chamber **100** below the gas distribution (showerhead) assembly **130**. The substrate support pedestal assembly **148** holds the substrate **103** during processing. The substrate support pedestal assembly **148** generally includes a plurality of lift pins (not shown) disposed therethrough that are configured to lift the substrate **103** from the substrate support pedestal assembly **148** and facilitate exchange of the substrate **103** with a robot (not shown) in a conventional manner. An inner liner **118** may closely circumscribe the periphery of the substrate support pedestal assembly **148**.

(20) In one implementation, the substrate support pedestal assembly **148** includes a mounting plate **162**, a base **164** and an electrostatic chuck **166**. The mounting plate **162** is coupled to the bottom **110** of the chamber body **102** includes passages for routing utilities, such as fluids, power lines and sensor leads, among others, to the base **164** and the electrostatic chuck **166**. The electrostatic chuck **166** comprises at least one clamping electrode **180** for retaining the substrate **103** below showerhead assembly **130**. The electrostatic chuck **166** is driven by a chucking power source **182** to develop an electrostatic force that holds the substrate **103** to the chuck surface, as is conventionally known. Alternatively, the substrate **103** may be retained to the substrate support pedestal assembly **148** by clamping, vacuum or gravity.

(21) At least one of the base **164** or electrostatic chuck **166** may include at least one optional embedded heater **176**, at least one optional embedded isolator **174** and a plurality of conduits **168**, **170** to control the lateral temperature profile of the substrate support pedestal assembly **148**. The conduits **168**, **170** are fluidly coupled to a fluid source **172** that circulates a temperature regulating fluid therethrough. The heater **176** is regulated by a power source **178**. The conduits **168**, **170** and heater **176** are utilized to control the temperature of the base **164**, thereby heating and/or cooling the electrostatic chuck **166** and ultimately, the temperature profile of the substrate **103** disposed thereon. The temperature of the electrostatic chuck **166** and the base **164** may be monitored using a plurality of temperature sensors **190**, **192**. The electrostatic chuck **166** may further comprise a plurality of gas passages (not shown), such as grooves, that are formed in a substrate support pedestal supporting surface of the chuck **166** and fluidly coupled to a source of a heat transfer (or backside) gas, such as He. In operation, the backside gas is provided at controlled pressure into the

gas passages to enhance the heat transfer between the electrostatic chuck **166** and the substrate **103**.

(22) In one implementation, the substrate support pedestal assembly **148** is configured as a cathode and includes an electrode **180** that is coupled to a plurality of RF power bias sources **184**, **186**. The RF bias power sources **184**, **186** are coupled between the electrode **180** disposed in the substrate support pedestal assembly **148** and another electrode, such as the showerhead assembly **130** or ceiling (lid **104**) of the chamber body **102**. The RF bias power excites and sustains a plasma discharge formed from the gases disposed in the processing region of the chamber body **102**.

(23) In the example depicted in FIG. **1**, the dual RF bias power sources **184**, **186** are coupled to the electrode **180** disposed in the substrate support pedestal assembly **148** through a matching circuit **188**. The signal generated by the RF bias power sources **184**, **186** is delivered through matching circuit **188** to the substrate support pedestal assembly **148** through a single feed to ionize the gas mixture provided in the plasma processing chamber **100**, thereby providing ion energy necessary for performing a deposition or other plasma enhanced process. The RF bias power sources **184**, **186** are generally capable of producing an RF signal having a frequency of from about 50 kHz to about 200 MHz and a power between about 0 Watts and about 5000 Watts. An additional bias power source **189** may be coupled to the electrode **180** to control the characteristics of the plasma.

(24) In one mode of operation, the substrate **103** is disposed on the substrate support pedestal assembly **148** in the plasma processing chamber **100**. A process gas and/or gas mixture is introduced into the chamber body **102** through the showerhead assembly **130** from the gas panel **158**. A vacuum pump system **128** maintains the pressure inside the chamber body **102** while removing deposition by-products.

(25) A controller **150** is coupled to the processing chamber **100** to control operation of the processing chamber **100**. The controller **150** includes a central processing unit (CPU) **152**, a memory **154**, and a support circuit **156** utilized to control the process sequence and regulate the gas flows from the gas panel **158**. The CPU **152** may be any form of general purpose computer processor that may be used in an industrial setting. The software routines can be stored in the memory **154**, such as random access memory, read only memory, floppy, or hard disk drive, or other form of digital storage. The support circuit **156** is conventionally coupled to the CPU **152** and may include cache, clock circuits, input/output systems, power supplies, and the like. Bi-directional communications between the controller **150** and the various components of the processing chamber **100** are handled through numerous signal cables.

(26) FIG. **2** illustrates a schematic view of a processing chamber **200** according to one embodiment. The processing chamber **200** may be used to process the substrate **103**, including providing thermal/heat energy to material layers disposed on the substrate **103**. The processing chamber **200** may include an array of radiant heating lamps **202** for heating, among other components, a back side **204** of a susceptor **220** disposed within walls **201** of the processing chamber **200**, and the substrate **103**. The susceptor **220** is supported by a susceptor support **218**. In the embodiment shown in FIG. **2**, the susceptor **220** has a ring shaped body with a central opening **203** and a lip **221** that extends from the edge of the susceptor **220** and circumscribes the central opening **203**. The lip **221** and the front side **202** of the susceptor **220** create a pocket **226** that supports the substrate **103** from the edge of the substrate to facilitate exposure of the substrate **103** to the thermal radiation provided by the lamps **202**.

(27) The susceptor **220** is located within the processing chamber **200** between an upper dome **210** and a lower dome **212**. The upper dome **210** is coupled to the lower dome **212** by a base ring **214**. The upper dome **210**, the lower dome **212** and the base ring **214** generally define an internal region of the processing chamber **200**. In some embodiments, the array of radiant heating lamps **202** may be disposed over the upper dome **210**. The substrate **301** can be brought into the processing chamber **200** and positioned onto the susceptor **220** through a loading port (not shown) formed in the base ring **214**.

(28) The susceptor **220** is shown in an elevated processing position, but may be moved vertically

by an actuator (not shown) to a loading position below the processing position to allow lift pins **222** to pass through holes in the susceptor support **218**, and raise the substrate **103** from the susceptor **220**. A robot (not shown) may then enter the processing chamber **200** to engage and remove the substrate **103** therefrom through the loading port. The susceptor **220** then may be actuated up to the processing position to place the substrate **103**, with a device side **224** facing up, on a front side **202** of the susceptor **220**.

(29) The susceptor **220** and the susceptor support **218**, while located in the processing position, divide the internal volume of the processing chamber **200** into a process gas region **228** that is above the substrate **103**, and a purge gas region **230** below the susceptor **220** and the susceptor support **218**. The susceptor **220** and susceptor support **218** are rotated during processing by a supporting cylindrical central shaft **232**, to minimize the effect of thermal and process gas flow spatial anomalies within the processing chamber **200**, and thus facilitate uniform processing of the substrate **103**. The central shaft **232** moves the substrate **301** in an up and down direction **234** during loading and unloading, and in some instances, during processing of the substrate **103**.

(30) In general, the central window portion of the upper dome **210** and the bottom of the lower dome **212** are formed from an optically transparent material, such as quartz. One or more lamps, such as an array of the lamps **202**, can be disposed adjacent to and beneath the lower dome **212** in a specified, optimal desired manner around the central shaft **232** to independently control the temperature at various regions of the substrate **103**. The heated substrate **103** is exposed to the process gases, thereby facilitating the thermal processing (i.e., deposition) of a material onto the upper surface of the substrate **103**.

(31) The lamps **202** may be configured to include bulbs **236** and may be configured to heat the substrate **301** to a temperature within a range of about 200 degrees Celsius to about 1600 degrees Celsius, for example, between about 300 degrees Celsius to about 1200 degrees Celsius or between about 500 to about 580 degrees Celsius. Each lamp **202** is coupled to a power distribution board (not shown) through which power is supplied to each lamp **202**. The lamps **202** are positioned within a lamphead **238** which may be cooled during or after processing by, for example, a cooling fluid introduced into channels **252** located between the lamps **202**. The lamphead **238** conductively and radiantly cools the lower dome **212** due in part to the close proximity of the lamphead **238** to the lower dome **212**. The lamphead **238** may also cool the lamp walls and walls of the reflectors (not shown) around the lamps. Alternatively, the lower dome **212** may be cooled by a convective approach known, such as by fans or other cooling techniques. Depending upon the application, the lampheads **238** may or may not be in contact with the lower dome **212**. As a result of backside heating of the substrate **103**, the use of an optical pyrometer **242** for temperature measurements/control on the substrate **103** and the susceptor **220** may also be utilized.

(32) A reflector **244** may be optionally placed outside the upper dome **210** to reflect infrared light that is radiating off the substrate **301** back onto the substrate **301**. The reflector **244** may be fabricated from a metal such as aluminum or stainless steel. The efficiency of the reflection can be improved by coating a reflector area with a highly reflective coating, such as with gold. The reflector **244** can have one or more machined channels **246** connected to a cooling source (not shown). The channel **246** connects to a passage (not shown) formed on a side of the reflector **244**. The passage is configured to carry a flow of a fluid such as water and may run horizontally along the side of the reflector **244** in any desired pattern covering a portion or entire surface of the reflector **244** for cooling the reflector **244**.

(33) Process gas supplied from a process gas supply source **248** is introduced into the process gas region **228** through a process gas inlet **250** formed in the sidewall of the base ring **214**. The process gas inlet **250** is configured to direct the process gas in a generally radially inward direction. During the film formation process, the susceptor **220** may be located in the processing position, which is adjacent to and at about the same elevation as the process gas inlet **250**, allowing the process gas to flow up and round along a flow path across the upper surface of the substrate **103** in a laminar flow.

The process gas exits the process gas region **228** through a gas outlet **255** located on the side of the processing chamber **200** opposite the process gas inlet **250**. Removal of the process gas through the gas outlet **255** may be facilitated by a vacuum pump **256** coupled thereto. As the process gas inlet **250** and the gas outlet **255** are aligned and disposed approximately at the same elevation, it is believed that such a parallel arrangement, when combined with a flatter upper dome **210** provides generally planar, uniform gas flow across the substrate **103**. Further radial uniformity may be provided by the rotation of the substrate **103** by the susceptor **220**.

(34) Purge gas may be supplied from a purge gas source **258** to the purge gas region **230** through an optional purge gas inlet **260** (or through the process gas inlet **250**) formed in the sidewall of the base ring **214**. The purge gas inlet **260** is disposed at an elevation below the process gas inlet **250**. The purge gas inlet **260** is configured to direct the purge gas in a generally radially inward direction. During the process, the susceptor **220** may be located at a position such that the purge gas flows down and along a flow path across the back side **204** of the susceptor **220** in a laminar flow. Without being bound by any particular theory, the flowing of the purge gas is believed to prevent or substantially avoid the flow of the process gas from entering into the purge gas region **230**, or to reduce diffusion of the process gas entering the purge gas region **230** (i.e., the region under the susceptor **220**). The purge gas exits the purge gas region **230** and is exhausted out of the processing chamber **200** through the gas outlet **255**, which is located on the side of the processing chamber **200** opposite the purge gas inlet **260**.

(35) FIG. 3 depicts a flow diagram illustrating a process **300** for manufacturing MTJ structures on a substrate for MRAM applications according to one embodiment of the present disclosure. FIGS. 4A-4E are schematic cross-sectional views of a substrate **400** at various stages of the process of FIG. 3. The process **300** is configured to be performed in a plasma processing chamber and a thermal processing chamber, such as the processing chambers **100**, **200** as described in FIGS. 1 and 2. It is contemplated that the process **300** may be performed in other suitable plasma immersion ion implantation systems or etching chamber, including those from other manufacturers.

(36) The process **300** begins at operation **302** by providing a substrate, such as the substrate **400** having a film stack **402** disposed thereon. FIG. 5 is a schematic diagram of one embodiment of a portion of the film stack **402** illustrated in FIG. 4A. In one embodiment, the substrate **400** comprises metal or glass, silicon, dielectric bulk material and metal alloys or composite glass, crystalline silicon (e.g., Si<100> or Si<111>), silicon oxide, strained silicon, silicon germanium, germanium, doped or undoped polysilicon, doped or undoped silicon wafers and patterned or non-patterned wafers silicon on insulator (SOI), carbon doped silicon oxides, silicon nitride, doped silicon, germanium, gallium arsenide, glass, or sapphire. The substrate **400** may have various dimensions, such as 200 mm, 300 mm, 450 mm or other diameter, as well as, being a rectangular or square panel. Unless otherwise noted, examples described herein are conducted on substrates with a 200 mm diameter, a 300 mm diameter, or a 450 mm diameter substrate. In one embodiment, the substrate **400**, as shown in FIG. 4A, includes the film stack **402** disposed on the substrate **400**.

(37) In one example, the film stack **402** includes a pinned magnetic layer **404**, an optional structure decoupling layer **405** and a tunneling barrier layer **408**, which may be an oxide barrier layer in the case of a tunnel junction magnetoresistive (TMR) sensor or a conductive layer in the case of a giant magnetoresistive (GMR) sensor, disposed above the pinned magnetic layer **404** sandwiched between a magnetic storage layer **410** and a magnetic reference layer **406**. When the film stack **402** is configured to form a TMR sensor, then the tunneling barrier layer **408** may comprise MgO, HfO₂, TiO₂, TaO_x, Al₂O₃, or other suitable materials. In the embodiment depicted in FIG. 5, the tunneling barrier layer **408** may comprise MgO having a thickness of about 1 to about 15 Angstroms, such about 10 Angstroms. The tunneling barrier layer **408** may be annealed either during or post deposition, for example, using a rapid thermal anneal (RTP) process.

(38) If the film stack **402** is configured to form a GMR sensor, film stack **402** may comprise a non-magnetic conductive material, such as copper. The structure decoupling layer **405** may be a metal

containing material or a magnetic material, such as Mo, Ta, W, CoFe, CoFeB and the like.

(39) In one example, the pinned magnetic layer **404** may have a thickness of about 2 nm. The pinned magnetic layer **404** may comprise one of several types of pinned layers, such as a simple pinned, antiparallel pinned, self pinned or antiferromagnetic pinned sensor. The pinned magnetic layer **404**, the magnetic storage layer **410** and the magnetic reference layer **406** may be constructed of several magnetic materials such as a metal alloy with dopants, such as boron dopants, oxygen dopants or other suitable materials. Metal alloys may be a nickel containing material, platinum containing material, Ru containing material, a cobalt containing material, tantalum containing materials and palladium containing materials. Suitable examples of the magnetic materials that may comprise the pinned magnetic layer **404** include Ru, Ta, Co, Pt, TaN, NiFeO.sub.x, NiFeB, CoFeO.sub.xB, CoFeB, CoFe, NiO.sub.xB, CoBO.sub.x, FeBO.sub.x, CoFeNiB, CoPt, CoPd, TaO.sub.x and the like.

(40) In the embodiment depicted in FIG. 4A, the magnetic storage layer **410** may be a CoFeB, CoFeNiB, Ta, Mo or W, combinations thereof or other suitable layer(s). For example, in the embodiment depicted in FIG. 5, the magnetic storage layer **410** includes a first CoFeB layer **508** and a second CoFeB layer **512** sandwiching an intermediate layer **510**. The first and second CoFeB layers **508**, **512** may have a thickness of about 5 to about 20 Angstroms, for example about 10 Angstroms, a composition of B in the range of about 10 to about 30 percent, and a composition of Fe in the range of about 20 to about 60 percent. The intermediate layer **510** may include one or more layers of at least one or more of Ta, Mo and W. The intermediate layer **510** may have a thickness of about 0 to about 10 Angstroms, for example about 3 Angstroms.

(41) The magnetic reference layer **406** may be CoFeB or other suitable layer. The magnetic reference layer **406** may have a thickness of about 5 to about 20 Angstroms, for example about 10 Angstroms. The magnetic reference layer **406** may have a composition of about 10 to about 30 percent of boron (B), and about 20 to about 60 percent Fe.

(42) The structure decoupling layer **405** is disposed between the magnetic reference layer **406** and the pinned magnetic layer **404**. The structure decoupling layer **405** may include one or more layers of at least one or more of Ta, Mo and W. The structure decoupling layer **405** may have a thickness of about 0 to about 10 Angstroms, for example about 3 Angstroms.

(43) The pinned magnetic layer **404** may include one or more layers. In the example depicted in FIG. 5, the pinned magnetic layer **404** sequentially includes a Co layer **520**, a Co/Pt layer **522**, an Ru layer **524**, a Co layer **526**, a Co/Pt layer **528**, a Pt layer **530**, a Ta layer **532**, an Ru layer **534**, a Ta layer **536** and a TaN layer **538**. The Co layer **520** may have a thickness of about 0 to about 10 Angstroms, for example about 5 Angstroms. The Ru layer **524** may have a thickness of about 3 to about 10 Angstroms, for example about 4-5 or about 7-9 Angstroms. The Co layer **526** may have a thickness of about 0 to about 10 Angstroms, for example about 5 Angstroms. The Pt layer **530**, the Ta layer **532**, the Ru layer **534**, the Ta layer **536** and the TaN layer **538** each may have a thickness of about 0 to about 5 Angstroms, for example about 3 Angstroms.

(44) The Co/Pt layer **522** may have a composition comprising [Co(.sub.x1)/Pt(.sub.y1)].sub.m, with x.sub.1 having a thickness of Co of between about 1 to about 6 Angstroms, y.sub.1 having a thickness of Pt of between about 1 Angstroms to about 10 Angstroms, for example about 1-8 Angstroms, and m having a number of between 0 and 5, where m represents the numbers of the Co/Pt layer **522** repeatedly formed in the film stack. For example, when x.sub.1 is 5 Angstroms and y.sub.1 is 3 Angstroms and m is number 2, it represents a film stack of Co layer (5 Angstroms)/Pt layer (3 Angstroms)/Co layer (5 Angstroms)/Pt layer (3 Angstroms).

(45) The Co/Pt layer **528** may have a composition comprising [Co(x)/Pt(y)]n, with x having a thickness of about 1 to about 6 Angstroms, y having a thickness of about 1 to about 10 Angstroms, for example about 1-8 Angstroms, and n having a number of between 3 and 10.

(46) A capping layer **412** is then formed on the magnetic storage layer **410** in the film stack **402**. In one embodiment, the capping layer **412** may include a magnetic layer. Suitable examples of the

capping layer **412** include one or more layers of at one or more of CoFeB, MgO, Ta and Ru. In the example depicted in FIG. 5, the capping layer **412** includes a first layer **502** and a second layer **506** sandwiching an intermediate layer **504**. The first layer **502** may have a thickness of about 0 to about 5 nanometers (nm), for example about 2 nm. The first layer **502** may include one or more layers comprised of Ta and/or Ru. The second layer **506** may comprise MgO, and have a thickness of about 0 to about 15 Angstroms, for example 7 Angstroms. The intermediate layer **504** may include CoFeB. The intermediate layer **504** may have a thickness of about 0 to about 50 Angstroms, for example about 10 Angstroms. The intermediate layer **504** may have a composition of about 10 to about 30 percent B, and about 20 to about 60 percent Fe.

(47) It is noted that the pinned magnetic layer **404**, the optional structure decoupling layer **405**, the tunneling barrier layer **408**, the magnetic storage layer **410**, the magnetic reference layer **406** and the capping layer **412** may be formed by any suitable techniques, such as CVD, PVD, ALD, spin-coating, spray coating, and any suitable manners. One example of systems that may be used to form these layers includes CENTURA®, PRECISION 5000® and PRODUCER® deposition systems, all available from Applied Materials Inc., Santa Clara, California. It is contemplated that other processing system, including those available from other manufacturers, may be adapted to practice the disclosure.

(48) At operation **304**, a patterning process, e.g., an etching process, is performed to remove a portion **407** of the film stack **402** exposed and defined by an etching mask layer (not shown) from the substrate **301**, until the underlying substrate **400** is exposed, as shown in FIG. 4B. The patterning process for patterning the film stack **402** may include several steps or different recipes configured to supply different gas mixtures or etchants to etch different layers in accordance with the materials included in each layer.

(49) During patterning, an etching gas mixture or several gas mixtures with different etching species are sequentially supplied into the substrate surface to remove the portion of the film stack **402** from the substrate **400**.

(50) The end point of the patterning process at operation **304** may be controlled by time or other suitable methods. For example, the patterning process may be terminated after performing for between about 200 seconds and about 10 minutes until the substrate **400** is exposed, as shown in FIG. 4B. The patterning process may be terminated by determination from an endpoint detector, such as an OES detector or other suitable detector as needed.

(51) At operation **306**, after the patterning process, a sidewall passivation layer **414** is formed on sidewalls **415** of the patterned film stack **402**, such as the sidewalls of the pinned magnetic layer **404**, the optional structure decoupling layer **405**, the tunneling barrier layer **408**, the magnetic storage layer **410**, the magnetic reference layer **406** and the capping layer **412**.

(52) In one example, the sidewall passivation layer **414** may be a dielectric layer formed by a deposition process performed after the patterning process at operation **304**. For example, the sidewall passivation layer **414** may be formed by a CVD deposition, PVD deposition, ALD deposition, or other suitable deposition processes as needed. The sidewall passivation layer **414** may be formed from a nitrogen containing material, carbon containing material, or oxide containing material. Suitable exemplary materials for forming the sidewall passivation layer **414** includes SiN, SiCN, SiO₂, SiON, SiC, amorphous carbon, SiOC, aluminum oxide (Al₂O₃), aluminum nitride (AlN), and the like.

(53) In another example, the sidewall passivation layer **414** may be formed from by-products generated during the patterning process at operation **304** accumulating on the sidewalls of the film stack **402** after the patterning process at operation **304**. For example, during patterning, a carbon containing gas, an oxygen containing gas, or a nitrogen containing gas may be utilized in the etching gas mixture during the patterning process. The carbon elements, oxygen elements and the nitrogen elements from these gases may react with elements from the film stack **402**, forming side-products, residuals or build-ups adhering on the sidewalls **415** of the film stack **402**. Thus, in the

example wherein the sidewall passivation layer **414** are formed from the patterning process from operation **304**, the gases supplied in the etching gas mixture at operation **304** may include at least one of carbon containing gas, oxygen containing gas, or nitrogen containing gas.

(54) After the sidewall passivation layer **414** is formed on the sidewalls **415** of the film stack **402**, a further deposition process may be performed to form an insulation layer **416** on the portion **407** of the substrate **400** where the film stack **402** was removed during the patterning process at operation **304**, as shown in FIG. **4C**. The insulation layer **416** may be any suitable insulating materials that may later under series of etching and deposition processes to form interconnection structure (e.g., a backend process) in the insulation layer **416** to complete the device structure fabrication process. In one example, the insulation layer **416** is a silicon oxide layer or other suitable material.

(55) At operation **308**, after the patterning process at operation **304** is completed and the sidewall passivation layer and the insulation layer **416** is formed on the substrate **400**, a thermal annealing process may be performed. The thermal annealing process is performed to repair, densify and enhance lattice structures of the film stack **402**, particularly the magnetic storage layer **410** and the magnetic reference layer **406** included in the film stack **402**. For example, after the thermal/annealing process, the magnetic storage layer **410** and the magnetic reference layer **406** may have stronger crystalline structures of (111), (220) and (311) plane peaks under XRD analysis prior to, e.g., without, the thermal/annealing process, thus enhancing crystallinity of the magnetic storage layer **410** and the magnetic reference layer **406** formed therein. During the annealing process, dopants from the magnetic storage layer **410** and the magnetic reference layer **406** are laterally driven away from the magnetic storage layer **410** and the magnetic reference layer **406**, as shown by the arrow **420**, to the sidewall passivation layer **414**, as shown in FIG. **4D**. By forming a sidewall passivation layer **414** between the insulation layer **416** and the film stack **402**, the dopants from the magnetic storage layer **410** and the magnetic reference layer **406** may be efficiently retained laterally proximate to the sidewall passivation layer **414**, rather than vertically moving towards the capping layer **412**, the tunneling barrier layer **408** or the pinned magnetic layer **404** or the optional structure decoupling layer **405** formed in the film stack **402**. Lateral diffusion, rather than vertical diffusion, of the dopants may efficiently control the dopants from drifting to the sidewall passivation layer **414** so that the film properties and crystallinity of the layers in the film stack **402** will not be adversely impacted or altered. For example, in the embodiment wherein boron (B) dopants are included in the magnetic storage layer **410** and the magnetic reference layer **406**, during the annealing process, boron dopants may be laterally diffused to the sidewall passivation layer **414**, forming boron compounds (such as boron oxide, boron nitride, or boron carbide, depending on the materials utilized to form the sidewall passivation layer **414**), which is a relatively stable compound that would not significantly alter electrical performance of the devices. Undesired dopant diffusion to the magnetic storage layer **410** and the magnetic reference layer **406** may result in with inaccurate magnetization or undesired atom spin directions when applying a magnetic field, thus adversely affecting the overall performance of the MTJ devices.

(56) Thus, by utilizing the thermal annealing process performed after the patterning process and/or after the sidewall passivation and/or insulation layer deposition process, a more stable magnetization process and a better control of the dopant diffusion process may be obtained, as compared to the conventional practices having the annealing process performed right after each layer is formed in the film stack or right after the film stack **402** is formed on the substrate **400**.

(57) In one example, the annealing process may be performed after the patterning process at operation **304** without or prior to forming the sidewall passivation layer **414** and the insulation layer **416**. In another example, the annealing process may be performed after the patterning process at operation **304** and the sidewall passivation and/or insulation layer deposition process at operation **306** as needed.

(58) The thermal annealing process may be performed in a thermal annealing chamber, such as the processing chamber **200** depicted in FIG. **2**. Alternatively, the annealing process may be performed

in any processing chamber configured to provide enough thermal energy to the film stack **402** disposed on the substrate **400**. In one embodiment, the thermal annealing process may be performed in-situ process in the processing chamber wherein the patterning process at operation **304** or the deposition process at operation **306** was performed. The thermal annealing process may heat the substrate **400** to a temperature greater than 200 degrees Celsius, such as between about 250 degrees Celsius and about 550 degree Celsius, to assist dopants in the magnetic storage layer **410** and the magnetic reference layer **406** to diffuse outward and laterally from the film stack and enhance crystallinity of the magnetic storage layer **410** and the magnetic reference layer **406**, and other layers in the film stack **402**.

(59) During annealing, an annealing gas mixture may be supplied. Gases that may be supplied in the annealing gas mixture may include a nitrogen containing gas, such as NH_3 , N_2 , NF_3 , N_2O , NO_2 , NO and the like, an inert gas, such as Ar, He, Ne, Kr, Xe or the like, or an oxygen containing gas, such as O_2 , O_3 , N_2O , H_2O , CO_2 , CO and the like.

(60) Alternatively, the annealing process may also be performed after the interconnection fabrication process (e.g., the back end fabrication process) is performed. It is noted that the interconnection fabrication process (e.g., the back end fabrication process) referred herein includes processes of patterning the insulation layer **416** to form trenches or vias in the insulation layer **416** and later having a conductive layer formed in trenches or vias defined in the insulation layer **416**. The temperature configured to perform the annealing process right after the deposition of the insulation layer **416** or sidewall passivation layer **414** deposition process at operation **306** and after the interconnection fabrication process (e.g., the back end fabrication process) may be the same or different. In one example, the temperature for the thermal annealing process configured to perform after the interconnection fabrication process (e.g., the back end fabrication process) may be slightly lower than the temperature of the thermal annealing process performed right after the deposition of the insulation layer **416** or sidewall passivation layer **414** deposition process at operation **306**.

(61) After the thermal annealing process, the magnetic storage layer **410** and the magnetic reference layer **406** may be turned into a crystallized magnetic storage layer **422** and a crystallized magnetic reference layer **424**, as shown in FIG. 4E, having crystal orientation mostly in one plane. The thermal energy provides during the thermal annealing process assists crystallizing the magnetic storage layer **410** and the magnetic reference layer **406** and other layers in the film stack **402** from amorphous state into crystalline state, thus efficiency enhancing magnetization of the film layers included in the film stack **402**. The thermal energy provided during the thermal annealing process at operation **308** assist growing the grains from the amorphous state into large sizes into crystallized grains, enhancing crystallinity of the magnetic storage layer **410** and the magnetic reference layer **406**, and other layers in the film stack **402**. As desired crystallization of the magnetic storage layer **410** and the magnetic reference layer **406** is obtained, the overall electrical properties of the film stack **402** for manufacturing MTJ devices are improved.

(62) Accordingly, processes and apparatus of forming MTJ device structures for MRAM are provided. The process advantageously provides a method to thermal anneal, e.g., thermal treat, a film stack utilized to form the MTJ devices. The thermal annealing process performed after the film stack deposition, patterning and sidewall passivation and insulation layer deposition process may efficiently crystallize the layers in the film stack in a manner (e.g., lateral diffusion rather than vertical diffusion) without damaging or impacting the film properties of the adjacent layers formed in the film stack. By doing so, the film stack utilized to form the MTJ structure may have a desired dopant profile and electrical performance and magnetization.

(63) While the foregoing is directed to embodiments of the disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof.

Claims

1. A method of forming a magnetic tunnel junction structure on a substrate, comprising: patterning a film stack having a tunneling barrier layer disposed between a magnetic reference layer and a magnetic storage layer disposed on a substrate to remove a portion of the film stack from the substrate until an upper surface of the substrate is exposed, wherein the magnetic storage layer comprises a first magnetic storage sublayer, a second magnetic storage sublayer and a metallic intermediate layer therebetween; forming a conformal sidewall passivation layer on sidewalls of the patterned film stack, wherein the sidewall passivation layer is in direct contact with a top surface of the substrate and covering the top surface, wherein the sidewall passivation layer has a vertical portion continuously coplanar along a vertical axis normal to a major axis of the substrate and a horizontal portion in direct contact with the substrate and extending outwardly from the vertical portion along the substrate; forming an insulation layer comprising silicon oxide, where the portion of the film stack was removed during patterning, that is in direct contact with the sidewall passivation layer such that the horizontal portion is between the insulation layer and the substrate and the insulation layer does not contact the top surface of the substrate; and subsequently performing a thermal annealing process to the film stack, wherein the insulation layer extends from a top surface of the sidewall passivation layer to the horizontal portion of the sidewall passivation layer wherein the film stack further comprises a pinning layer disposed between the substrate and the magnetic reference layer, the pinning layer further comprising sequentially from the magnetic reference layer in the direction of the substrate; a Co layer, a Co/Pt layer, an Ru layer, a Co layer, a Co/Pt layer, a Pt layer, a Ta layer, an Ru layer, a Ta layer, and a TaN layer.
2. The method of claim 1, wherein performing the thermal annealing process further comprises: maintaining the substrate temperature at between about 250 degrees Celsius and about 550 degree Celsius.
3. The method of claim 1, wherein the film stack further comprises a capping layer disposed on the magnetic storage layer.
4. The method of claim 1, wherein the film stack further comprises a pinning layer disposed between the substrate and the magnetic reference layer.
5. The method of claim 4, wherein the film stack further comprises a structure decoupling layer disposed between the pinning layer and the magnetic reference layer.
6. A method of forming a magnetic tunnel junction structure on a substrate, comprising: patterning a film stack having a tunneling barrier layer disposed between a magnetic reference layer and a magnetic storage layer disposed on a substrate to remove a portion of the film stack from the substrate until an upper surface of the substrate is exposed, wherein the magnetic storage layer comprises a first magnetic storage sublayer, a second magnetic storage sublayer and a metallic intermediate layer therebetween; forming a sidewall passivation layer comprising carbon containing materials or nitrogen containing materials on sidewalls of the patterned film stack, the sidewall passivation layer having a vertical portion coplanar along a vertical axis normal to a major axis of the substrate and a horizontal portion extending outwardly from the vertical portion along the substrate such that the horizontal portion covers the upper surface of the substrate where the portion of the film stack was removed during patterning; and subsequently performing a thermal annealing process to the film stack wherein the film stack further comprises a pinning layer disposed between the substrate and the magnetic reference layer, the pinning layer further comprising sequentially from the magnetic reference layer in the direction of the substrate; a Co layer, a Co/Pt layer, an Ru layer, a Co layer, a Co/Pt layer, a Pt layer, a Ta layer, an Ru layer, a Ta layer, and a TaN layer.
7. The method of claim 6, wherein performing the thermal annealing process further comprises: maintaining the substrate temperature at between about 250 degrees Celsius and about 550 degree Celsius.

Celsius.

8. The method of claim 6, wherein the film stack further comprises a capping layer disposed on the magnetic storage layer.
 9. The method of claim 6, wherein the film stack further comprises a pinning layer disposed between the substrate and the magnetic reference layer.
 10. The method of claim 9, wherein the film stack further comprises a structure decoupling layer disposed between the pinning layer and the magnetic reference layer.
 11. The method of claim 6, wherein the magnetic reference layer and the magnetic storage layer are fabricated from metal alloy with boron dopants.
 12. The method of claim 6, wherein forming the sidewall passivation layer further comprises: forming a dielectric layer on the sidewalls of the patterned film stack by a chemical vapor deposition process.
 13. The method of claim 12, wherein the dielectric layer of the sidewall passivation layer is at least one of SiN, SiCN, SiO₂, SION, SiC, amorphous carbon, amorphous carbon, SiOC, or aluminum nitride (AlN).
 14. The method of claim 6, wherein the sidewall passivation layer is formed from by-products generated during the patterning of the film stack.
 15. The method of claim 6, wherein forming the sidewall passivation layer further comprises: forming an insulation layer on the substrate where the portion of the film stack is removed.
 16. The method of claim 15, wherein the thermal annealing process is performed after a back end process is performed in the insulation layer or right after the insulation layer is formed on the substrate.
 17. The method of claim 6, wherein performing the thermal annealing process further comprises: altering crystalline structures of the magnetic reference layer and a magnetic storage layer.
 18. The method of claim 6, wherein performing the thermal annealing process further comprises: driving dopants in the magnetic reference layer and the magnetic storage layer laterally outward to the sidewall passivation layer.
 19. The method of claim 18, wherein the dopant reacts with the sidewall passivation layer to form a dopant compound.
 20. A method of forming a magnetic tunnel junction structure on a substrate, comprising: patterning a film stack having a tunneling barrier layer disposed between a magnetic reference layer and a magnetic storage layer disposed on a substrate to remove a portion of the film stack from the substrate until an upper surface of the substrate is exposed, wherein the magnetic storage layer comprises a first magnetic storage sublayer, a second magnetic storage sublayer and a metallic intermediate layer therebetween; forming a sidewall passivation layer on sidewalls of the patterned film stack, the sidewall passivation layer having a vertical portion coplanar along a vertical axis normal to a major axis of the substrate and a horizontal portion extending outwardly from the vertical portion along the substrate such that the sidewall passivation layer covers the top surface of the substrate; forming an insulation layer comprising silicon oxide, where the portion of the film stack was removed during patterning, that is in direct contact with the sidewall passivation layer on the substrate such that the horizontal portion is between the insulation layer and the substrate such that the insulation layer does not contact the top surface of the substrate; and driving dopants from the magnetic reference layer and the magnetic storage layer laterally outward into the sidewall passivation layer by a thermal treatment process wherein the film stack further comprises a pinning layer disposed between the substrate and the magnetic reference layer, the pinning layer further comprising sequentially from the magnetic reference layer in the direction of the substrate; a Co layer a Co/Pt layer, an Ru layer, a Co layer, a Co/Pt layer, a Pt layer, a Ta layer, an Ru layer, a Ta layer, and a TaN layer.
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